Reg. No. | | | | | | | | | | |

[12EE62/13EE62]

SIXTH SEMESTER B.E/B.TECH EXAMINATION, JUNE - 2016

DIGITAL SIGNAL PROCESSING

Time:3 Hours

Maximum Marks:70

Instructions:

1. Answer any **FIVE** full Questions.
2. Normalised filter tables are allowed.

1. a. Show that with $x(n)$ as an N-point sequence and $X(k)$ as its N-point DFT $Y(k) = X(k)e^{\frac{-j\pi mk}{N}}$ when y(n)=x((n-m))$_N$ **04**

b. If the DFT $\{x(n)\} = x(k) = \{4, -j2, 0, j2\}$ using the properties of DFT, find **06**

i. DFT of $x(n-2)$

ii. DFT of $x^2(n)$

c. Given $x(n) = \{1, -2, 3, 0, -1, 1\}$ evaluate the following functions without computing DFT. **04**

i. $x((n-2))_6$

ii. $x((n+3))_6$

2. a. The impulse response of a linear system is $\{1, -2, 2\}$. Determine the output response of the system when the input is $\{1, 2, 1,\}$ use circular convolution. **06**

b. A long sequence is filtered through a filter with impulse response $h(n) = \{1, -1\}$. Compute the output y(n) of the filter using overlap add method. Use 4-point circular convolution input $x(n) = \{1, -2, 2, -1, 3, -4, 4, -3, 2\}$ **08**

3. a. Develop decimation in time FFT algorithm considering N=8 **06**

b. Compute the 8-point DFT of the sequence $x(n) = \{0.5, 0.5, 0.5, 0.5, 0, 0, 0, 0\}$ use DIT .FFT **08**

4. a. Why FFT needed? What is the speed improvement factor in calculating 64 point DFT of a sequence using direct computation & FFT algorithm? **05**

b. First five points of eight point DFT of a real valued sequence is given by $X(k) = \{0, 2+j2, -j4, 2-j2, 0\}$. Determine the remaining points and find the sequence $x(n)$ using DIF-FFT algorithm. **09**

5. a. A system functions is specified by its transfer functions $H(Z)=\frac{0.28Z^2+0.319Z+0.04}{0.5Z^3+0.3Z^2+0.17Z-0.2}$ realise the system by direct form-I and direct form II **08**

b. Realise the system functions $H(Z)=1-\frac{3}{8}Z^{-1}-\frac{3}{32}Z^{-2}+\frac{5}{64}Z^{-3}-\frac{3}{16}Z^{-4}$ in **06**

i. Direct form

ii. Cascade form.

6. a. The transfer functions of an analog filter is $H(S)=\frac{1}{(S+1)(S+2)}$. Determine H(Z) using impulse invariant transformation, **04**

b. Determine $H(Z)$ for a lowest order bulter worth filter satisfying following constraints. $0.8 \le |H(e^{jw})| \le 1 \quad 0 \le |w| \le 0.2\pi \quad |H(e^{jw})| \le 0.2 \quad 0.6\pi \le |w| \le \pi$ **10**

Take T= 1 sec. Apply impulse invariant transformations.

7. a. For the analog transfer functions $H(S)=\frac{2}{(S+1)(S+3)}$ determine $H(Z)$ using Bilinear transformations take T=2 sec. **04**

b. Design an IIR low pass Bulterworth digital filter to satisfy the following analog specifications. Pass band ripple=1dB **10**

Pass band edge frequency=100 π rad/sec

Stop band attenuations = 35db

Stop band edge frequency = 1000 π rad/sec

Sampling rate of 2000 samples/sec. use bilinear transformation technique.

8. a. Design an FIR low pass filter using rectangular window with the following desired frequency response **07**

$Hd(e^{jw})=e^{-j(N-1)w/2}$ for $-\frac{\pi}{4} \le |w| \le \frac{\pi}{4}$ and $Hd(e^{jw})=0$ for $\frac{\pi}{4} \le |w| \le \pi$ Take filter length as 5.

b. Determine impulse response h(n) of a filter having the derived frequency response **07**

$Hd(e^{jw})=e^{-j3w}$ for $0 \le |w| \le \frac{\pi}{4}$ and $Hd(e^{jw})=0$ for $\frac{\pi}{2} \le |w| < \pi$ take N=7 use frequency sampling approach.